

JAMIE FULFORD

✉ jamiehf2@illinois.edu

🔗 jfulfo.github.io

🌐 linkedin.com/in/jamie-fulford

🐙 github.com/jfulfo

Education

University of Illinois Urbana-Champaign

August 2025 – Present

Ph.D. Computer Science — Advised by Talia Ringer

Champaign, Illinois

- **Research:** proof autoformalization, programming languages, formal methods, interactive theorem proving

University of Virginia

August 2022 – May 2025

B.S. Computer Science and Mathematics

Charlottesville, Virginia

- **GPA:** 3.9

Research Experience

University of Illinois Urbana-Champaign

August 2025 – Present

Graduate Research Assistant

Champaign, Illinois

- **DARPA expMath:** building tooling for **proof autoformalization** and **proof decomposition** for **Agda**, targeting homotopy type theory — decomposing a theorem statement into multiple lemmas dispatched to automated provers
- **Proof translation between program logics:** designing languages with specific interpretable properties, then implementing them in **Rocq** to interoperate and compile proofs between the two

Virginia Tech Systems Software Research Group

January 2025 – July 2025

Research Assistant (Proof Engineer)

Remote

- Developed type systems and compiler verification for **Linux eBPF** in **Rocq**, proving kernel safety properties
- Extended the verified compiler framework **CompCert** to a new frontend language

University of Pennsylvania

May 2024 – August 2024

Research Fellow, NSF REU (advised by Steve Zdancewic)

Philadelphia, Pennsylvania

- Designed and formalized a novel lambda calculus for parallel and distributed computation in **Rocq**, with machine-checked proofs of **type safety** and **confluence**

Software & Formalizations

Formalizing LLM Guardrails with AgentSpec in Lean 4 🐙

Fall 2025

- Sole author. Executable Lean 4 semantics for **AgentSpec**, a DSL for runtime enforcement of LLM agent behavior, parameterized over predicate tables and external effects
- 42 machine-checked theorems, no sorry: **safety is compositional in both directions**, the safety evaluator correctly implements the violation semantics (no false positives or misattribution), per-enforcement contracts, and termination

ProbaLean: A Probabilistic DSL in Lean 4 🐙

Spring 2026

- Joint, three authors (equal contribution). Discrete probabilistic language in Lean 4 with **denotational semantics over s-finite measures** on Mathlib, plus executable operational semantics
- Proved the interpretation is a **monad morphism** into the Giry monad, **commutativity of independent sampling**, and **soundness** of denotational against operational semantics — axiom-free at $\mathbb{Q}_{\geq 0}$

Teaching

Head TA, Computer Systems & Organization, UVA (500+ students)

Fall 2023, 2024

Awards

PLMW Scholarship — Programming Languages Mentoring Workshop, POPL 2025

January 2025

Technical Skills

Proof Assistants: Agda, Rocq (Coq), Lean 4

Verification: Iris, CompCert, Mathlib

Languages: OCaml, Haskell, Rust, Python, C, C++, Java, JavaScript, SQL, Bash

Tools: Git, Linux, NixOS, L^AT_EX, PyTorch